

Signed uniformly distributed sequences from ordinary ones

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Source Question

The source paper is Mercourakis and Vassiliadis, *Uniform Distribution of Sequences and its interplay with Functional Analysis*, arXiv:2102.02306. Definition 1 of that paper says that, for a compact Hausdorff space K and a real signed regular Borel measure $\mu \in M(K)$ with $\|\mu\| = |\mu|(K) = 1$, a sequence $(x_n) \subseteq K$ is μ -u.d. if

1. (x_n) is $|\mu|$ -uniformly distributed in the classical sense, and
2. there are signs $\varepsilon_n \in \{\pm 1\}$ such that

$$\frac{1}{N} \sum_{n=1}^N \varepsilon_n \delta_{x_n} \xrightarrow{w^*} \mu.$$

The source proves this property for compact metric K . In concluding future-work item 2(b), it asks about compact non-metrizable spaces K for which every probability measure $\nu \in P(K)$ admits an ordinary u.d. sequence:

For such spaces it would be interesting to know if every signed measure μ with $\|\mu\| = 1$ admits a u.d. sequence in the sense of Definition 1.

for examples (and the properties) of functions of Volterra type, we refer the reader to the books [11], pp.35-36 and [6], pp.22-25 and 33-35.

2. Concerning future work, we note the following:

(a) It would be interesting to have a generalization of Theorems 7 and 8 for functions of bounded variation of several variables, that is, for functions f defined on the cube I^n for $n \geq 2$, which satisfy a proper notion of bounded variation (see for instance Definition 5.2, p.147 of [15]).

(b) Besides compact metric spaces (see Theorem 3), there are several classes of compact non-metrizable spaces K with the property that every measure $\mu \in P(K)$ admits a u.d. sequence (see [16] and [18]). For such spaces it would be interesting to know if every signed measure μ with $\|\mu\| = 1$ admits a u.d. sequence in the sense of Definition 1 (cf. also Remark 5(b)). In our opinion the most interesting case is that of compact separable groups G ; since we know that under special set-theoretic assumptions, i.e. Continuum Hypothesis (CH) (see [16]) or Martin's axiom plus the negation of Continuum Hypothesis ($\text{MA} + \neg \text{CH}$), (see [8]), every measure $\mu \in P(G)$ admits a u.d. sequence. We note that it is enough to consider the compact group $\{0, 1\}^c$, where c = the cardinality of the continuum; this is so because, as is well known, every compact separable group is a dyadic space, i.e. a continuous image of $\{0, 1\}^c$ (see [12] and [18]).

(c) Regarding our consideration, the case of locally compact and separable metrizable spaces is also of interest, see [15] Notes, pp.177-178.

Answer

Theorem 1. *Let K be a compact Hausdorff space with the following property: every probability measure $\nu \in P(K)$ admits a uniformly distributed sequence in the classical sense. Then every real signed regular Borel measure $\mu \in M(K)$ with $\|\mu\| = 1$ admits a μ -u.d. sequence in the sense of Definition 1 of the source paper.*

Proof Intuition

The only obstruction in the signed setting is that the same points must simultaneously realize the total variation $|\mu|$ and, after signs are attached, realize μ . The Jordan decomposition separates these two tasks. Choose an ordinary uniformly distributed sequence for the normalized positive variation and another for the normalized negative variation. Then merge the two lists so that their asymptotic proportions are exactly the two Jordan masses. Testing against any $f \in C(K)$, the unsigned Cesaro averages converge to the weighted sum $\mu^+ + \mu^- = |\mu|$, while the signed averages converge to $\mu^+ - \mu^- = \mu$.

Proof

Let $\mu = \mu^+ - \mu^-$ be the Jordan decomposition of μ , and set

$$\alpha = \mu^+(K), \quad \beta = \mu^-(K).$$

Since $\|\mu\| = |\mu|(K) = \mu^+(K) + \mu^-(K)$, we have

$$\alpha + \beta = 1.$$

If $\alpha = 1$, then $\mu = |\mu|$ is a probability measure. By the hypothesis on K , choose a μ -u.d. sequence in the ordinary sense and take all signs equal to $+1$. This is a μ -u.d. sequence in the source's signed sense. If $\beta = 1$, choose an ordinary $|\mu|$ -u.d. sequence and take all signs equal to -1 . Thus assume from now on that $\alpha, \beta > 0$.

By the hypothesis on K , choose sequences $(a_i) \subseteq K$ and $(b_j) \subseteq K$ such that

$$\frac{1}{m} \sum_{i=1}^m \delta_{a_i} \xrightarrow{w^*} \frac{\mu^+}{\alpha}, \quad \frac{1}{m} \sum_{j=1}^m \delta_{b_j} \xrightarrow{w^*} \frac{\mu^-}{\beta}.$$

Choose a subset $A \subseteq \mathbb{N}$ with asymptotic density α . For example,

$$A = \{n \in \mathbb{N} : \lfloor n\alpha \rfloor > \lfloor (n-1)\alpha \rfloor\}.$$

Then, with $A_N = A \cap \{1, \dots, N\}$, one has

$$|A_N| = \lfloor N\alpha \rfloor, \quad \frac{|A_N|}{N} \rightarrow \alpha.$$

Let $B = \mathbb{N} \setminus A$; then $|B \cap \{1, \dots, N\}|/N \rightarrow \beta$.

Define a sequence $(x_n) \subseteq K$ and signs (ε_n) as follows. If n is the i -th element of A , put

$$x_n = a_i, \quad \varepsilon_n = +1.$$

If n is the j -th element of B , put

$$x_n = b_j, \quad \varepsilon_n = -1.$$

Fix $f \in C(K)$. Write $p_N = |A_N|$ and $q_N = N - p_N$. Since $\alpha, \beta > 0$, both p_N and q_N tend to infinity. Therefore

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^N f(x_n) &= \frac{p_N}{N} \left(\frac{1}{p_N} \sum_{i=1}^{p_N} f(a_i) \right) + \frac{q_N}{N} \left(\frac{1}{q_N} \sum_{j=1}^{q_N} f(b_j) \right) \\ &\longrightarrow \alpha \int_K f d\frac{\mu^+}{\alpha} + \beta \int_K f d\frac{\mu^-}{\beta} = \int_K f d|\mu|. \end{aligned}$$

Thus (x_n) is $|\mu|$ -u.d. in the classical sense.

Similarly,

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^N \varepsilon_n f(x_n) &= \frac{p_N}{N} \left(\frac{1}{p_N} \sum_{i=1}^{p_N} f(a_i) \right) - \frac{q_N}{N} \left(\frac{1}{q_N} \sum_{j=1}^{q_N} f(b_j) \right) \\ &\longrightarrow \alpha \int_K f d\frac{\mu^+}{\alpha} - \beta \int_K f d\frac{\mu^-}{\beta} = \int_K f d\mu. \end{aligned}$$

Equivalently,

$$\frac{1}{N} \sum_{n=1}^N \varepsilon_n \delta_{x_n} \xrightarrow{w^*} \mu.$$

The two defining conditions from Definition 1 are satisfied, so (x_n) is μ -u.d. This proves the theorem.

□

Consequences for the Source Question

Theorem 1 answers future-work item 2(b) affirmatively for every compact Hausdorff space K satisfying the probability-measure hypothesis stated in that item. In particular, whenever an external theorem gives that all probability measures on one of the non-metrizable compact spaces mentioned there admit ordinary u.d. sequences, the signed-measure conclusion follows immediately by the theorem above.

The argument does not address the separate future-work item 2(a), concerning functions of bounded variation of several variables, nor the locally compact noncompact direction in item 2(c).

Verification Notes

No numerical or computer-assisted verification is involved. The proof reduces the signed problem to two ordinary probability-measure u.d. sequences and a deterministic density partition of $\mathbb{N}$. The weak-star convergence is checked directly on arbitrary $f \in C(K)$.

Novelty Check

Before promotion, the run's cheap indexes were searched for arXiv:2102.02306, the title, the authors, and core phrases including “signed measure u.d. sequence”, “signed measure uniformly distributed”, and “compact non-metrizable signed measure”. No prior packet, attempt, or proof-gap entry for this target was found.

Local corpus searches found the source paper and unrelated signed-sum/vector sequence material, but no later solution of the future-work item. Web/arXiv spot checks on 2026-06-30 for exact phrases around “2102.02306 signed measure uniformly distributed”, “every signed measure admits a u.d. sequence”, “compact non-metrizable signed measure u.d. sequence”, and “Mercourakis Vassiliadis Uniform Distribution of Sequences” found the source paper but no exact later resolution.

Human Review Recommendation

The main review point is whether the source future-work item 2(b) is intended exactly under the hypothesis that every probability measure on K admits an ordinary u.d. sequence. Under that reading, the proof is a complete affirmative answer. The density interleaving calculation is the only substantive step to check.

References

- [1] S. K. Mercourakis and G. Vassiliadis, *Uniform Distribution of Sequences and its interplay with Functional Analysis*, arXiv:2102.02306.